

Collage Theorem :

Let $\{f_1, f_2, \dots, f_m\}$ be an IFS on a complete metric space (X, d) , i.e., $d(f_i(x), f_i(y)) \leq c_i d(x, y)$ $\forall x, y \in X$,

where $0 < c_i < 1$. Let E be a nonempty compact subset of X . Then

$$D(E, A) \leq \frac{1}{1-c} D(E, \bigcup_{i=1}^m f_i(E)),$$

where A is the attractor of the IFS,

$$c = \max_{1 \leq i \leq m} c_i.$$

and D is the Hausdorff distance.

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Proof : We have $A = \bigcup_{i=1}^m f_i(A)$.

$$\begin{aligned} D(E, A) &\leq D(E, \bigcup_{i=1}^m f_i(E)) + D(\bigcup_{i=1}^m f_i(E), A) \\ &= D(E, \bigcup_{i=1}^m f_i(E)) + D(\bigcup_{i=1}^m f_i(E), \bigcup_{i=1}^m f_i(A)) \\ &\leq D(E, \bigcup_{i=1}^m f_i(E)) + c D(E, A) \end{aligned}$$

$$\Rightarrow (1-c) D(E, A) \leq D(E, \bigcup_{i=1}^m f_i(E))$$

$$\Rightarrow D(E, A) \leq \frac{1}{1-c} D(E, \bigcup_{i=1}^m f_i(E))$$

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Corollary: Let E be a nonempty compact subset of $\mathbb{R}^n$. Given $\delta > 0$, there exists an IFS of similarities $\{S_1, S_2, \dots, S_m\}$ with attractor A satisfying $D(E, A) < \delta$.

Proof: Let $B_1, B_2, \dots, B_m$ be a collection of balls that cover E and which have centres in E and radii at most $\delta/4$. (This can be done because E is compact).

$$\Rightarrow E \subseteq \bigcup_{i=1}^m B_i \subseteq N_{\delta/4}(E) \quad \text{--- (i)}$$

For each i , let S_i be a contracting similarity of ratio less than $\frac{1}{2}$ that map E into B_i .

Then $S_i(E) \subseteq B_i \subseteq N_{\delta/4}(S_i(E))$ for each i .

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$$\Rightarrow \bigcup_{i=1}^m S_i(E) \subseteq \bigcup_{i=1}^m B_i \subseteq N_{\delta/4}(E) \quad (\text{from (i)})$$

$$\text{Also, } E \subseteq \bigcup_{i=1}^m B_i \subseteq \bigcup_{i=1}^m N_{\delta/2}(S_i(E)) \subseteq N_{\delta/2}\left(\bigcup_{i=1}^m S_i(E)\right)$$

$\therefore$ By the definition of the Hausdorff metric,

$$D(E, \bigcup_{i=1}^m S_i(E)) \leq \delta/2$$

$\therefore$ By the collage theorem, $\bigcup_{i=1}^m S_i(E)$

$$D(E, A) \leq \frac{1}{1-c} D(E, \bigcup_{i=1}^m S_i(E)) < 2 \cdot \frac{\delta}{2} = \delta \quad (\because c < \frac{1}{2})$$

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Fractal Interpolation

What is an interpolation?

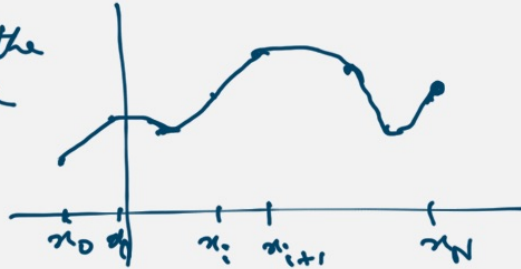
Suppose $x_0 < x_1 < x_2 < \dots < x_N$ be real numbers and consider the dataset $\{(x_i, y_i) : 0 \leq i \leq N\}$ in $\mathbb{R}^2$

An interpolation for the given dataset is a continuous function

$$f : [x_0, x_N] \rightarrow \mathbb{R}$$

such that $f(x_i) = y_i$

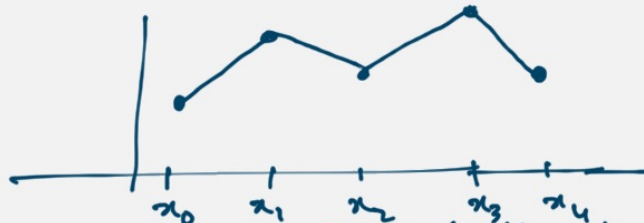
for $i = 0, 1, 2, \dots, N$.



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Examples:

① Piecewise linear interpolation:



Continuous but not differentiable at x_i 's.

② Polynomial interpolation

$$\{(x_i, y_i) : 0 \leq i \leq N\}$$

There exists a polynomial $p(x)$ of degree N

$$\text{s.t. } p(x_i) = y_i \quad \forall i = 0, 1, \dots, N.$$

• Infinitely differentiable

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What is a fractal interpolation?

Let a set of data $\{(x_i, F_i) : i=0, 1, \dots, N\}$ be given, where $x_0 < x_1 < x_2 < \dots < x_N$.

A fractal interpolation is a continuous function $f : [x_0, x_N] \rightarrow \mathbb{R}$ such that

(i) $f(x_i) = F_i$ for $i=0, 1, 2, \dots, N$.

(ii) The graph of f , $G(f) = \{(x, f(x)) : x_0 \leq x \leq x_N\}$ is the attractor of some IFS.

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Piecewise linear interpolation

$f : [x_0, x_N] \rightarrow \mathbb{R}$ is given by

$$f(x) = F_{i-1} + \left(\frac{x - x_{i-1}}{x_i - x_{i-1}} \right) (F_i - F_{i-1})$$

for $x \in [x_{i-1}, x_i]$
 $i \in \{1, 2, \dots, N\}$

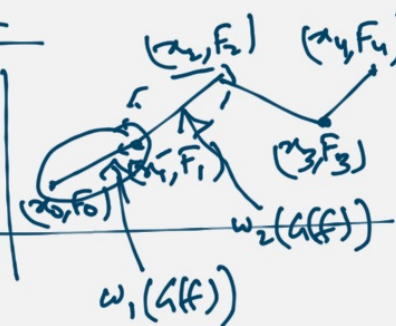
We define an IFS given by

$$w_n : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix}$$

for $n=1, 2, \dots, N$.

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We want $G(f) = \bigcup_{n=1}^N w_n(G(f))$

$$\text{ie. } \left\{ \begin{pmatrix} x \\ f(x) \end{pmatrix} \right\} = \bigcup_{n=1}^N \left\{ \begin{pmatrix} a_n & 0 \\ c_n & 0 \end{pmatrix} \begin{pmatrix} x \\ f(x) \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix} \right\}$$

$$x \in [x_0, x_N]$$

$$w_n \begin{pmatrix} x \\ f(x) \end{pmatrix} = \begin{pmatrix} a_n x + e_n \\ c_n x + f_n \end{pmatrix}$$

We want :

$$\begin{aligned} a_n x_0 + e_n &= x_{n-1} \\ a_n x_N + e_n &= x_n \\ c_n x_0 + f_n &= F_{n-1} \\ c_n x_N + f_n &= F_n \end{aligned}$$

$$\Rightarrow a_n = \frac{x_n - x_{n-1}}{x_N - x_0} ; e_n = \frac{x_N x_{n-1} - x_0 x_n}{x_N - x_0}$$

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$$c_n = \frac{F_n - F_{n-1}}{x_N - x_0} ; f_n = \frac{x_N F_{n-1} - x_0 F_n}{x_N - x_0}$$

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